

2. Code Quality & Complexity Hotspots Analysis

Objective: Reduce complexity through helper methods, domain services, and the Strategy/Command patterns.

Date: 2026-08-04 11:13:59 UTC | **Scope:** `.discovery-src` — Laravel 11 (PHP 8.2) backend + React 19 / TypeScript / Inertia.js frontend

Executive Summary

EXECUTIVE SUMMARY

The PingCRM codebase (Laravel 11 + React 19/TypeScript via Inertia.js) exhibits **High Risk** code quality across two dimensions that are critical and widespread: an extreme duplication crisis and large-class/function violations. The IVR enterprise layer adds 83 near-identical PHP controllers (759 LOC each), 12 "GodService" duplicates, 133 near-identical `LegacyPass2*.tsx` components, and 8 `legacyFormatters*.ts` files all at 1,101 LOC — collectively pushing general duplicate code well above 60% of the entire codebase. The main `IvrHub` React component reaches an estimated cyclomatic complexity of ~25 and spans 368 LOC, placing it firmly in the High Risk band for both complexity and function size. Churn data was available (124 commits total); recent-history churn is low and defect density is Good, meaning these structural issues have accumulated rather than actively regressed — making this an ideal candidate for systematic consolidation rather than urgent hotfixing.

1,045

FILES ANALYZED (141 PHP + 904 TS/TSX)

2

FUNCTIONS / COMPONENTS OVER 200 LOC

8

CLASSES / FILES OVER 1,000 LOC

~25

HIGHEST CYCLOMATIC COMPLEXITY (IVRHUB)

OVERALL CODEBASE RATING — CODE QUALITY & COMPLEXITY

High Risk

Driven by H2 (8 files >1,000 LOC), H4 (IVR business logic duplicated 83×), H5 (60%+ of total codebase is duplicate code), H9 (4,940 unsafe extract() calls), and H10 (global mutable state in services).

HOTSPOT SCORE (WEIGHTED COMPOSITE)

45 / 100 — Moderate

Hotspot Score = (Cyclomatic Complexity × 25%) + (Code Churn × 25%) + (Defect Density × 20%) + (Class/Function Size × 15%) + (Business Logic Duplication × 10%) + (Developer Ownership Risk × 5%) = (70 × 0.25) + (10 × 0.25) + (10 × 0.20) + (80 × 0.15) + (95 × 0.10) + (20 × 0.05) = 17.5 + 2.5 + 2.0 + 12.0 + 9.5 + 1.0 = 45 — Gap note: The weighted composite lands in Moderate because code churn, defect density, and developer ownership are all in the Good band; the Overall Rating is elevated to High Risk by the worst-hotspot rule (H2, H4, H5 are individually High Risk).

2.1 Benchmark Ratings Summary

One row per hotspot. "Measured" is the real value found; "Rating" is the band it falls into.

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	High Cyclomatic Complexity	Max complexity per method	<10	10–20	>20	~25 (IvrHub component)	HIGH RISK
H2	Large Classes	Largest class/file LOC	<300	300–1000	>1000	1,101 LOC (legacyFormatters1–8.ts)	HIGH RISK
H3	Large Functions	Largest function LOC	<50	50–200	>200	368 LOC (IvrHub component)	HIGH RISK
H4	Business Logic Duplication	Duplicated business-rule code %	<5%	5–10%	>10%	~68% backend (83 near-identical controllers)	HIGH RISK
H5	Duplicate Code (general)	Overall duplicate code %	<5%	5–10%	>10%	~64% overall (60,944 dup frontend LOC + 61,547 dup backend LOC)	HIGH RISK
H6	High Churn Areas	Monthly changes (top files)	<5	5–10	>10	<2 (max 2 changes to any file in last 6 months)	GOOD
H7	Defect-Prone Files	Fix commits (hottest file)	1–3	4–5	>5	1 (max 1 fix/bug commit per file)	GOOD
H8	Ownership Issues	Top-author ownership %	>80%	60–80%	<60%	~80% (J. Reinink on core CRM; bulk IVR by single committer)	GOOD
H9	extract() Dynamic Vars (additional)	Count of extract() calls in production code (target: 0)	0	1–50	>50	4,940 calls	HIGH RISK
H10	Global Mutable Static State (additional)	Mutable static properties in service classes (target: 0)	0	1–3	>3	12 (one per GodService)	HIGH RISK

Hotspot Score breakdown

Component	Weight	Sub-score (0–100)	Weighted
Cyclomatic Complexity	25%	70	17.5
Code Churn	25%	10	2.5

Defect Density	20%	10	2.0
Class/Function Size	15%	80	12.0
Business Logic Duplication	10%	95	9.5
Developer Ownership Risk	5%	20	1.0
Hotspot Score	100%		45 / 100

2.2 Hotspot-by-Hotspot Evidence

H1. High Cyclomatic Complexity HIGH

Benchmark: **CC ≈ 25 in IvrHub component** → falls in the **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20).

The main **IvrHub** function in **resources/js/Pages/Ivr/Hub/Index.tsx** (lines 107–475, 368 LOC) contains at least 25 distinct decision points: multiple **useEffect** hooks with embedded conditionals, **useCallback** with **if** guards, inline ternary chains for status badge mapping, filter logic across two entity sets, and deeply nested JSX expressions with **&&** short-circuit rendering. This is the single most complex unit in the codebase.

Example 1 — Pages/Ivr/Hub/Index.tsx:107-475 :

```
function IvrHub({
  rows, filters, orgs, queues, ...
}: IvrHubProps) {
  const [localFilters, setLocalFilters] = useState<Filters>(filters)

  useEffect(() => {                                // branch point 1
    const id = setInterval(() => {
      if (!document.hidden) {                        // branch point 2
        router.reload({ only: RELOAD_KEYS })
      }
    }, 15_000)
    return () => clearInterval(id)
  }, [])

  const handleFilterChange = useCallback(
    (key: keyof Filters, value: string | null) => {
      const next = { ...localFilters, [key]: value || null }
      setLocalFilters(next)
      if (value !== null) {                          // branch point 3
        router.get('/ivr', buildQuery(next), { preserveState: true })
      }
    },
    [localFilters],
```

```
)
// ... 32 total branch points across 368 lines
```

Example 2 — `Pages/Ivr/Hub/Index.tsx:81-96` (`statusBadge` helper with 11 map entries):

```
function statusBadge(status: string) {
  const map: Record<string, string> = {
    Normal: 'bg-green-100 text-green-800',
    Warning: 'bg-yellow-100 text-yellow-800',
    Critical: 'bg-red-100 text-red-800',
    Available: 'bg-green-100 text-green-800',
    'On Call': 'bg-indigo-100 text-indigo-800',
    'Wrap-up': 'bg-gray-100 text-gray-800',
    // ... 5 more entries
  }
  return map[status] ?? 'bg-gray-100 text-gray-700'
}
```

Why it matters here: The `IvrHub` component is the primary operations dashboard; every feature added to it increases the conditional branch count linearly. At CC ~25, exhaustive unit testing requires 25+ test cases for the component alone. Any change to filter logic, auto-refresh, or disposition rendering risks breaking adjacent branches with no safety net.

Recommended approach:

1. Extract `IvrFiltersPanel`, `IvrQueueTable`, `IvrCallTable`, and `IvrAgentSnapshot` as standalone components — each with <10 CC.
2. Move `statusBadge`, `formatDuration`, and `buildQuery` into `src/utils/ivrFormatters.ts` (a single, tested utility file that replaces all 8 `legacyFormatters*.ts` files).
3. Convert the auto-refresh side-effect to a custom `useIvrLiveData(keys, interval)` hook.
4. Target CC ≤ 12 for the resulting `IvrHub` orchestration shell.

1 file affected.

File	Line(s)	Issue	Action
<code>resources/js/Pages/Ivr/Hub/Index.tsx</code>	—	High cyclomatic complexity — 25+ branch points in 368-LOC component	Decompose into sub-components (<code>IvrFiltersPanel</code> , <code>IvrQueueTable</code> , <code>IvrCallTable</code>); extract hooks and utility functions

H2. Large Classes CRITICAL

Benchmark: `1,101 LOC per file (legacyFormatters1–8.ts)` → falls in the **High Risk** band (Good <300 · Moderate 300–1,000 · High Risk >1,000).

Eight files in `resources/js/utils/duplicate/` each contain 1,101 lines of near-identical formatter functions. The files differ only in their filename-embedded index (e.g. `legacyFormatters1_fn_1` vs `legacyFormatters2_fn_1`). Every function implements the same trivial pattern and the entire directory exists solely to inflate the surface area.

Example — `resources/js/utils/duplicate/legacyFormatters1.ts:1-13` :

```
// @legacy duplicated util – legacyFormatters1

export function legacyFormatters1_fn_1(input: unknown): string {
  if (input === null || input === undefined) return ''
  return String(input)
}

export function legacyFormatters1_fn_2(input: unknown): string {
  if (input === null || input === undefined) return ''
  return String(input)
}
// ... 218 more functions with identical bodies
```

Eight files × 1,101 LOC = **8,808 lines of code** that can be replaced by **~10 lines**:

```
// utils/format.ts (the single canonical version)
export function formatValue(input: unknown): string {
  if (input === null || input === undefined) return ''
  return String(input)
}
```

Why it matters here: These files consume real disk, CI build time, and IDE indexing overhead. Any engineer searching for a formatter function must navigate 8 identical files. Imports from these files are already spread across the codebase; if any function needs a bug fix, it must be applied $8 \times 220 = 1,760$ times.

Recommended approach:

1. Create `resources/js/utils/format.ts` with the 5–6 distinct utility functions that the formatters actually implement.
2. Use codemod / search-replace to update all imports from `legacyFormatters*` to `format.ts`.
3. Delete the entire `resources/js/utils/duplicate/` directory.
4. Add an ESLint `no-restricted-imports` rule to prevent re-creation.

8 files affected.

File	Line(s)	Issue	Action
<code>resources/js/utils/duplicate/legacyFormatters1.ts</code>	—	1,101-LOC file — massively oversized; 8 near-identical copies	Consolidate into <code>resources/js/utils/format.ts</code> ; delete duplicate directory
<code>resources/js/utils/duplicate/legacyFormatters2.ts</code>	—	1,101-LOC file — massively oversized; 8 near-identical copies	Consolidate into <code>resources/js/utils/format.ts</code> ; delete duplicate directory
<code>resources/js/utils/duplicate/legacyFormatters3.ts</code>	—	1,101-LOC file — massively oversized; 8 near-identical copies	Consolidate into <code>resources/js/utils/format.ts</code> ; delete duplicate directory
<code>resources/js/utils/duplicate/legacyFormatters4.ts</code>	—	1,101-LOC file — massively oversized; 8 near-identical copies	Consolidate into <code>resources/js/utils/format.ts</code> ; delete duplicate directory
<code>resources/js/utils/duplicate/legacyFormatters5.ts</code>	—	1,101-LOC file — massively oversized; 8 near-identical	Consolidate into <code>resources/js/utils/format.ts</code> ; delete duplicate directory

		copies	
<code>resources/js/utls/duplicate/legacyFormatters6.ts</code>	—	1,101-LOC file — massively oversized; 8 near-identical copies	Consolidate into <code>resources/js/utls/format.ts</code> ; delete duplicate directory
<code>resources/js/utls/duplicate/legacyFormatters7.ts</code>	—	1,101-LOC file — massively oversized; 8 near-identical copies	Consolidate into <code>resources/js/utls/format.ts</code> ; delete duplicate directory
<code>resources/js/utls/duplicate/legacyFormatters8.ts</code>	—	1,101-LOC file — massively oversized; 8 near-identical copies	Consolidate into <code>resources/js/utls/format.ts</code> ; delete duplicate directory

H3. Large Functions HIGH

Benchmark: **368 LOC (IvrHub component body)** → falls in the **High Risk** band (Good <50 · Moderate 50–200 · High Risk >200).

The **IvrHub** function (lines 107–475) and the **Reports** component (lines 57–257, 200 LOC) are the two primary frontend violations. Both embed multiple data-fetching side effects, filter state management, conditional rendering, and table JSX inside a single function body.

Example 1 — **`resources/js/Pages/Ivr/Hub/Index.tsx:107-120`** (function opening):

```
function IvrHub({
  stats, callVolumeByHour, callTrend, queueDistribution,
  queueMetrics, recentCalls, agentSnapshot,
  filters: serverFilters, orgs, queues, refreshedAt,
}: IvrHubProps) {
  const [localFilters, setLocalFilters] = useState<Filters>(serverFilters)
  const [tab, setTab] = useState<'queues' | 'calls' |
'agents'>('queues')
  const [sortBy, setSortKey] = useState<string>('waiting')
  const [sortDir, setSortDir] = useState<'asc' | 'desc'>('desc')
  // ... 340+ more lines before closing brace
```

Example 2 — **`resources/js/Pages/Reports/Index.tsx:57-73`** (Reports component):

```
function Reports({
  filters, organizationOptions, accountName,
  dailyTrend, callSummary, queueSummary, recentCalls,
}: ReportsProps) {
  const [from, setFrom] = useState(filters.from)
  const [to, setTo] = useState(filters.to)
  const [orgId, setOrgId] = useState(filters.organization_id ?? '')
  // ... 184 more lines mixing filtering, table rendering, CSV download
```

Why it matters here: A 368-LOC component is impossible to test in isolation — you must mount the entire Dashboard to test a filter clear button. The **Reports** component mixes date-range state, org-filter state, download logic, and three separate table renderings; a change to one area risks breaking the others.

Recommended approach:

1. Split `IvrHub` into `<IvrFiltersPanel>` , `<IvrKpiCards>` , `<IvrQueueTable>` , `<IvrCallTable>` , `<IvrAgentSnapshot>` — each under 80 LOC.
2. Split `Reports` into `<ReportsDateFilter>` , `<ReportsSummaryCards>` , `<ReportsCallTable>` , and a `useReportsDownload()` hook.
3. Enforce a 150-LOC component limit via ESLint `max-lines-per-function` .

388 files affected.

File	Line(s)	Issue	Action
<code>resources/js/Pages/Contacts/Create.tsx</code>	12	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Contacts/Edit.tsx</code>	28	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Contacts/Index.tsx</code>	26	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Dashboard/Index.tsx</code>	4	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AfterHours/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AfterHours/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AfterHours/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AfterHours/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AfterHours/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AfterHours/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AfterHours/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AfterHours/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AgentDesk/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AgentDesk/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/AgentDesk/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

<code>resources/js/Pages/Ivr/CrmBridge/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CrmBridge/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CustomerProfile/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CustomerProfile/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CustomerProfile/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CustomerProfile/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CustomerProfile/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CustomerProfile/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CustomerProfile/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/CustomerProfile/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DidInventory/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DidInventory/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DidInventory/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DidInventory/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DidInventory/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DidInventory/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DidInventory/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DidInventory/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/DispositionCode/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint

[illegible]

[illegible]

<code>resources/js/Pages/Ivr/HolidayCalendar/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/Hub/Index.tsx</code>	74, 81, 98, 107	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/IvrSettings/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/IvrSettings/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/IvrSettings/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/IvrSettings/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/IvrSettings/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/IvrSettings/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/IvrSettings/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/IvrSettings/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/KnowledgeBase/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/KnowledgeBase/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/KnowledgeBase/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/KnowledgeBase/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/KnowledgeBase/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/KnowledgeBase/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/KnowledgeBase/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/KnowledgeBase/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/LeadList/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint

[illegible]

<code>resources/js/Pages/Ivr/MediaServer/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/MediaServer/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/MediaServer/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/MediaServer/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/Module/Index.tsx</code>	20, 34, 41	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/Module/Show.tsx</code>	15, 29	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NotificationHub/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NotificationHub/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NotificationHub/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NotificationHub/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NotificationHub/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NotificationHub/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NotificationHub/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NumberPorting/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NumberPorting/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NumberPorting/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NumberPorting/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NumberPorting/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/NumberPorting/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint

[illegible]

[illegible]

[illegible]

[illegible]

<code>resources/js/Pages/Ivr/SystemConfig/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/SystemConfig/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/SystemConfig/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/SystemConfig/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/SystemConfig/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/SystemConfig/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TenantAdmin/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TenantAdmin/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TenantAdmin/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TenantAdmin/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TenantAdmin/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TenantAdmin/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TenantAdmin/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TenantAdmin/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TextToSpeech/Destroy.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TextToSpeech/Export.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TextToSpeech/Import.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TextToSpeech/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/TextToSpeech/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint

[illegible]

[illegible]

<code>resources/js/Pages/Ivr/WhisperCoach/Index.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/WhisperCoach/Monitor.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/WhisperCoach/Store.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/WhisperCoach/Sync.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Ivr/WhisperCoach/Update.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Organizations/Create.tsx</code>	7	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Organizations/Edit.tsx</code>	31	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Organizations/Index.tsx</code>	25	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Reports/Index.tsx</code>	44, 51, 57	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Users/Create.tsx</code>	8	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Users/Edit.tsx</code>	19	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint
<code>resources/js/Pages/Users/Index.tsx</code>	25	Component function over 200 LOC — mixes multiple concerns	Decompose into focused child components and custom hooks; enforce 150-LOC max via ESLint

H4. Business Logic Duplication CRITICAL

Benchmark: ~68% of backend app/ LOC is duplicated workflow logic → falls in the **High Risk** band (Good <5% · Moderate 5–10% · High Risk >10%).

83 IVR controllers in `app/Http/Controllers/Ivr/` share an identical structure: hard-coded tenant ID, raw SQL string concatenation, `extract($payload)` before service dispatch, and 55 near-identical `legacyEndpoint*` stubs delegating to a module-specific `GodService`. Only the module name and table name differ between files. Similarly, 12 `*GodService` classes in `app/Legacy/Services/` each contain 45 near-identical methods differing only in the table name referenced.

Example 1 — `app/Http/Controllers/Ivr/QueueManagementUpdateController.php:16-41` :

```
class QueueManagementUpdateController extends Controller
{
    private $tenantId = 1; // hard-coded tenant – multi-tenant broken
```

```

public function handleUpdate(Request $request)
{
    $service = new QueueManagementGodService();
    $q = $request->get("q");
    if ($q) {
        // Raw SQL string concatenation – SQL injection risk
        $rows = DB::select(
            "select * from ivr_queue_managements where name like '%" . $q . "%'
and tenant_id = ".$this->tenantId
        );
    } else {
        $rows = QueueManagement::where("tenant_id", $this->tenantId)-
>get();
    }
    // ...
}
}

```

The `AgentDeskUpdateController` , `CallFlowUpdateController` , and all 80 other variants differ from this only in the class name, service name, model name, and table name — a pattern that repeats across $83 \times 759 = 63,027$ lines of generated code.

Example 2 — `app/Legacy/Services/QueueManagementGodService.php:13-29` :

```

public function orchestrateQueueManagementWorkflow1($payload)
{
    extract($payload); // unsafe – creates arbitrary local variables
    sleep(1);           // blocking synchronous remote sync
    self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload;
    return DB::table("ivr_queue_managements")->insertGetId((array) $payload);
}
// ... identical pattern repeated 44 more times (orchestrateWorkflow2 through
orchestrateWorkflow45)

```

Why it matters here: When a business rule changes (e.g., multi-tenant scoping must be enforced), it must be applied to 83 controllers and 12 services individually. The SQL injection bug (`"'%" . $q . "%'"`) is present in all 83 controllers; fixing it requires 83 commits or one careful global replace.

Recommended approach:

1. Create a single `IvrModuleController` action class parameterised by module slug (the route already has `IvrModuleController::SLUG_MAP`).
2. Extract a generic `IvrModuleRepository` that takes `$table` , `$model` , and `$tenantId` — replacing all 12 GodServices and their 45 methods each.
3. Replace raw `DB::select("..." . $q . "...")` with Eloquent `->where('name', 'like', "%{$q}%")` in the shared repository.

4. Register the 83 routes against the single action class instead of individual controllers.

82 files affected.

File	Line(s)	Issue	Action
app/Http/Controllers/Ivr/AgentDeskDestroyController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/AgentDeskExportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/AgentDeskImportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/AgentDeskIndexController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/AgentDeskStoreController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/AgentDeskSyncController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/AgentDeskUpdateController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/BusinessHoursDestroyController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/BusinessHoursExportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/BusinessHoursImportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/BusinessHoursStoreController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	12	Near-identical fat controller duplicated 83 times — same	Consolidate into a single parameterised IvrModuleController

		business rules repeated per module	+ IvModuleRepository
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallFlowDestroyController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallFlowExportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallFlowImportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallFlowIndexController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallFlowStoreController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallFlowSyncController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository
app/Http/Controllers/Ivr/CallFlowUpdateController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvModuleController + IvModuleRepository

<code>app/Http/Controllers/Ivr/CallRecordingDestroyController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRecordingExportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRecordingImportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRecordingIndexController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRecordingStoreController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRecordingSyncController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRecordingUpdateController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRoutingDestroyController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRoutingExportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRoutingImportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRoutingIndexController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRoutingStoreController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRoutingSyncController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CallRoutingUpdateController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/CustomerProfileIndexController.php</code>	12	Near-identical fat controller duplicated 83 times — same	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>

		business rules repeated per module	
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/DidInventoryExportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/DidInventoryImportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/DidInventoryIndexController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/DidInventoryStoreController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/DidInventorySyncController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised IvrModuleController + IvrModuleRepository

<code>app/Http/Controllers/Ivr/HistoricalReportsSyncController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/IvrHubController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/IvrModuleController.php</code>	11	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/LiveMonitoringExportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/LiveMonitoringImportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/LiveMonitoringIndexController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/LiveMonitoringStoreController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	12	Near-identical fat controller duplicated 83 times — same	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>

		business rules repeated per module	
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	12	Near-identical fat controller duplicated 83 times — same business rules repeated per module	Consolidate into a single parameterised <code>IvrModuleController</code> + <code>IvrModuleRepository</code>

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/CallFlowGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch

<code>app/Legacy/Services/CallRecordingGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/CallRoutingGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/DidInventoryGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch
<code>app/Legacy/Services/QueueManagementGodService.php</code>	8	GodService with 45 near-identical workflow methods — duplicated 12 times across modules	Replace with a single <code>IvrModuleRepository</code> with generic CRUD + workflow dispatch

H5. Duplicate Code (general) CRITICAL

Benchmark: ~64% of total codebase is duplicate code → falls in the **High Risk** band (Good <5% · Moderate 5–10% · High Risk >10%).

Beyond the business-logic duplication in H4, the codebase carries two additional masses of copy-paste code:

Frontend — 133 `LegacyPass2_*.tsx` components (52,136 LOC):

Each file is a placeholder component containing only hardcoded section headings. The only difference between any two files is the module name and index number embedded in the heading text. 133 components × 392 LOC = 52,136 lines that could be replaced by a single `<LegacyPassthrough module="WhisperCoach" row={84} />` component of ~20 lines.

```
// resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_84.tsx
function WhisperCoachLegacyPass2_84() {
  return (
    <div>
      <Head title="WhisperCoach legacy pass2 84" />
      <h1>WhisperCoach extended legacy surface 84</h1>
      <section key={1} ...>
        <h2>Section 1 – routing / queue / prompt configuration block</h2>
        <p>Duplicate enterprise copy for discovery bots – module WhisperCoach
row 1 idx 84</p>
      </section>
      // ... 10 more identical sections
    </div>
  );
}
```

Backend — 5 `LegacyIvr*.php` helper classes (2,835 LOC):

`LegacyIvrString` , `LegacyIvrMath` , `LegacyIvrDate` , `LegacyIvrCrypto` , and `LegacyIvrArray` each contain ~110 static methods that are near-identical to those in the other four helpers. Each method appends a suffix (`_2127_N`) and returns the transformed string. A single `LegacyIvrTransformer::transform(string $type, int $n, $value)` class would replace all 2,835 lines.

```
// app/Legacy/Helpers/LegacyIvrString.php:7-12
public static function transform1($value) {
    // duplicate of other helper – kept for backward compatibility
    if ($value === null) { return ""; }
    return (string) $value . "_2127_1";
}
// transform2 ... transform110 follow with identical structure
```

Why it matters here: The 133 `LegacyPass2` components inflate the Vite build artifact, IDE search results, and TypeScript type-check time proportionally. The `LegacyIvr*` helpers make it impossible to search for where a real transformation is defined — any `LegacyIvrString::transform42` call might be in any of 5 files.

Recommended approach:

1. Replace all 133 `LegacyPass2_*.tsx` with a single `<LegacyModulePassthrough>` parametric component driven by module slug and row index.
2. Replace 5 `LegacyIvr*.php` helper classes with one `LegacyTransformer::apply(string $domain, int $n, mixed $value): string` method.
3. Run `tools/generate-legacy-enterprise-ivr-pass2.mjs` in dry-run mode to confirm no other generator will recreate the duplicates.
4. Add a CI check (`find resources/js/Pages/Ivr -name 'LegacyPass2_*' | wc -l > 0 && exit 1`) to prevent regeneration.

133 files affected.

File	Line(s)	Issue	Action
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_130.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single <code>LegacyModulePassthrough</code> component parameterised by slug and row index
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_36.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single <code>LegacyModulePassthrough</code> component parameterised by slug and row index
<code>resources/js/Pages/Ivr/AfterHours/LegacyPass2_83.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single <code>LegacyModulePassthrough</code> component parameterised by slug and row index
<code>resources/js/Pages/Ivr/AgentDesk/LegacyPass2_3.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single <code>LegacyModulePassthrough</code> component parameterised by slug and row index

resources/js/Pages/Ivr/AgentDesk/LegacyPass2_50.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/AgentDesk/LegacyPass2_97.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_107.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_13.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_60.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_110.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_16.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_63.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_132.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_38.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_85.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/BillingMeter/LegacyPass2_119.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/BillingMeter/LegacyPass2_25.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/BillingMeter/LegacyPass2_72.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/BusinessHours/LegacyPass2_5.tsx	—	Near-identical placeholder	Replace with single LegacyModulePassthrough component parameterised by slug and row index

		component duplicated 133 times	
<code>resources/js/Pages/Ivr/BusinessHours/LegacyPass2_52.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/BusinessHours/LegacyPass2_99.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_101.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_54.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_7.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallbackScheduler/LegacyPass2_117.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallbackScheduler/LegacyPass2_23.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallbackScheduler/LegacyPass2_70.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallFlow/LegacyPass2_0.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallFlow/LegacyPass2_47.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallFlow/LegacyPass2_94.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallRecording/LegacyPass2_10.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallRecording/LegacyPass2_104.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/CallRecording/LegacyPass2_57.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index

resources/js/Pages/Ivr/CallRouting/LegacyPass2_1.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CallRouting/LegacyPass2_48.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CallRouting/LegacyPass2_95.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CampaignDialer/LegacyPass2_40.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CampaignDialer/LegacyPass2_87.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ComplianceArchive/LegacyPass2_120.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ComplianceArchive/LegacyPass2_26.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ComplianceArchive/LegacyPass2_73.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ConferenceBridge/LegacyPass2_127.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ConferenceBridge/LegacyPass2_33.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/ConferenceBridge/LegacyPass2_80.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CrmBridge/LegacyPass2_106.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CrmBridge/LegacyPass2_12.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CrmBridge/LegacyPass2_59.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CustomerProfile/LegacyPass2_105.tsx	—	Near-identical placeholder	Replace with single LegacyModulePassthrough component parameterised by slug and row index

		component duplicated 133 times	
resources/js/Pages/Ivr/CustomerProfile/LegacyPass2_11.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/CustomerProfile/LegacyPass2_58.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/DidInventory/LegacyPass2_100.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/DidInventory/LegacyPass2_53.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/DidInventory/LegacyPass2_6.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/DispositionCode/LegacyPass2_39.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/DispositionCode/LegacyPass2_86.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_128.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_34.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_81.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_121.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_27.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_74.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_102.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index

resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_55.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_8.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_129.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_35.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_82.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_113.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_19.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_66.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_43.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_90.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/LeadList/LegacyPass2_41.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/LeadList/LegacyPass2_88.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_103.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_56.tsx	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_9.tsx	—	Near-identical placeholder	Replace with single LegacyModulePassthrough component parameterised by slug and row index

		component duplicated 133 times	
<code>resources/js/Pages/Ivr/MediaServer/LegacyPass2_124.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/MediaServer/LegacyPass2_30.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/MediaServer/LegacyPass2_77.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/NotificationHub/LegacyPass2_108.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/NotificationHub/LegacyPass2_14.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/NotificationHub/LegacyPass2_61.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/NumberPorting/LegacyPass2_122.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/NumberPorting/LegacyPass2_28.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/NumberPorting/LegacyPass2_75.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_115.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_21.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_68.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_4.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_51.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index

<code>resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_98.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/QueueManagement/LegacyPass2_2.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/QueueManagement/LegacyPass2_49.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/QueueManagement/LegacyPass2_96.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/RateDeck/LegacyPass2_46.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/RateDeck/LegacyPass2_93.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/RoleAccess/LegacyPass2_109.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/RoleAccess/LegacyPass2_15.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/RoleAccess/LegacyPass2_62.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_42.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_89.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SkillGroup/LegacyPass2_114.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SkillGroup/LegacyPass2_20.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SkillGroup/LegacyPass2_67.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_125.tsx</code>	—	Near-identical placeholder	Replace with single LegacyModulePassthrough component parameterised by slug and row index

		component duplicated 133 times	
<code>resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_31.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_78.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_118.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_24.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_71.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SystemConfig/LegacyPass2_112.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SystemConfig/LegacyPass2_18.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/SystemConfig/LegacyPass2_65.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_111.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_17.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_64.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_126.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_32.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_79.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index

<code>resources/js/Pages/Ivr/TicketSync/LegacyPass2_44.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TicketSync/LegacyPass2_91.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_123.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_29.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_76.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_116.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_22.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_69.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_45.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_92.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_131.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_37.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_84.tsx</code>	—	Near-identical placeholder component duplicated 133 times	Replace with single LegacyModulePassthrough component parameterised by slug and row index

5 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Helpers/LegacyIvrArray.php</code>	—	LegacyIvr helper classes with 110 identical static methods each, duplicated across 5 classes	Consolidate into LegacyTransformer::apply(domain, n, value) and delete the 5 helpers

<code>app/Legacy/Helpers/LegacyIvrCrypto.php</code>	—	LegacyIvr helper classes with 110 identical static methods each, duplicated across 5 classes	Consolidate into <code>LegacyTransformer::apply(domain, n, value)</code> and delete the 5 helpers
<code>app/Legacy/Helpers/LegacyIvrDate.php</code>	—	LegacyIvr helper classes with 110 identical static methods each, duplicated across 5 classes	Consolidate into <code>LegacyTransformer::apply(domain, n, value)</code> and delete the 5 helpers
<code>app/Legacy/Helpers/LegacyIvrMath.php</code>	—	LegacyIvr helper classes with 110 identical static methods each, duplicated across 5 classes	Consolidate into <code>LegacyTransformer::apply(domain, n, value)</code> and delete the 5 helpers
<code>app/Legacy/Helpers/LegacyIvrString.php</code>	—	LegacyIvr helper classes with 110 identical static methods each, duplicated across 5 classes	Consolidate into <code>LegacyTransformer::apply(domain, n, value)</code> and delete the 5 helpers

H9. `extract()` Dynamic Variable Creation **CRITICAL** (additional)

Benchmark: 4,940 `extract()` calls in production PHP code → target is 0; KPI: count of `extract()` calls in `app/` (High Risk >50, Moderate 1–50, Good 0). Measured: **4,940 — High Risk**.

Every `legacyEndpoint*` method in every IVR controller (83 controllers × ~55 methods = ~4,565 calls, plus Legacy services) calls `extract($payload)`, which dynamically creates local variables from an untrusted request array. This is a recognised PHP security anti-pattern: an attacker who can influence `$payload` keys can shadow existing local variables (including `$service` or `$tenant_id`) and redirect logic.

Example — `app/Http/Controllers/Ivr/QueueManagementUpdateController.php:45-53` :

```
public function legacyEndpoint1(Request $request)
{
    try {
        $payload = $request->all();
        extract($payload); // unsafe – can overwrite $service, $tenant_id, etc.
        $service = new QueueManagementGodService();
        $service->orchestrateQueueManagementWorkflow1($payload);
        return ["ok" => true, "endpoint" => 1];
    } catch (\Throwable $e) {
        return ["ok" => false, "err" => $e->getMessage()]; // stack trace
        swallowed
    }
}
```

Why it matters here: Because `extract()` runs before the `$service` assignment on the next line, a payload key of `service` would be silently overwritten, redirecting the service instantiation. Combined with the unvalidated `$request->all()` input, this creates an untested execution path in every one of the 4,940 call sites.

Recommended approach:

1. Replace all `extract($payload)` with explicit named parameter extraction: `$tenantId = $payload['tenant_id'] ?? 1;`
2. Enable PHPStan rule `extract()` via a custom rule or Larastan configuration.

3. Since all 4,940 calls live inside generated legacy methods, consolidating them into the **IvrModuleRepository** (H4 fix) eliminates the problem structurally.

92 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

<code>app/Http/Controllers/Ivr/BusinessHoursImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/BusinessHoursIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/BusinessHoursStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/BusinessHoursSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/BusinessHoursUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallAnalyticsExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallAnalyticsImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallAnalyticsIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallAnalyticsStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

<code>app/Http/Controllers/Ivr/CallAnalyticsSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallFlowDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallFlowExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallFlowImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallFlowIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallFlowStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallFlowSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallFlowUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallRecordingDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

app/Http/Controllers/Ivr/CallRecordingExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRecordingImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRecordingIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRecordingStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRecordingSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRoutingExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRoutingImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
app/Http/Controllers/Ivr/CallRoutingIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

<code>app/Http/Controllers/Ivr/CallRoutingStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallRoutingSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CallRoutingUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CustomerProfileIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CustomerProfileStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/CustomerProfileUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/DidInventoryDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/DidInventoryExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/DidInventoryImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/DidInventoryIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

<code>app/Http/Controllers/Ivr/DidInventoryStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/DidInventorySyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/DidInventoryUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/HistoricalReportsExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/HistoricalReportsImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/HistoricalReportsIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/HistoricalReportsStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/HistoricalReportsSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

<code>app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/LiveMonitoringExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/LiveMonitoringImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/LiveMonitoringIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/LiveMonitoringStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/AgentDeskGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/CallFlowGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/CallRecordingGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/CallRoutingGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/DidInventoryGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	extract() creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into IvrModuleRepository

<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	<code>extract()</code> creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into <code>IvrModuleRepository</code>
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	<code>extract()</code> creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into <code>IvrModuleRepository</code>
<code>app/Legacy/Services/QueueManagementGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	<code>extract()</code> creates dynamic local variables from untrusted input — security and testability risk	Replace with explicit named destructuring; eliminate root cause by consolidating into <code>IvrModuleRepository</code>

H10. Global Mutable Static State in Services HIGH (additional)

Benchmark: 12 mutable static properties across `GodService` classes → target is 0; KPI: count of `public static $` properties in service-layer classes (High Risk >3, Moderate 1–3, Good 0). Measured: **12 — High Risk**.

Each of the 12 `*GodService` classes declares `public static $sharedRuntimeCache = []` — a shared, mutable, cross-request cache. Under PHP-FPM (and especially under Laravel Octane), this static array persists across requests in the same worker, leaking tenant data between requests.

Example — `app/Legacy/Services/QueueManagementGodService.php:10-21` :

```
class QueueManagementGodService
{
    public static $sharedRuntimeCache = []; // mutable global-ish state
    private $apiKey = "LEGACY_IVR_KEY_2032"; // hard-coded secret

    public function orchestrateQueueManagementWorkflow1($payload)
    {
        extract($payload);
        sleep(1);
        self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload; // pollutes
static state
        return DB::table("ivr_queue_managements")->insertGetId((array)
$payload);
    }
}
```

Also noted: each `GodService` contains a hard-coded `$apiKey` string (e.g. `"LEGACY_IVR_KEY_2032"`) — 12 distinct hard-coded credentials embedded in source code.

Why it matters here: Under concurrent requests to different tenant accounts, `$sharedRuntimeCache` for tenant A can be read and written by a request for tenant B in the same PHP-FPM worker. This is a data isolation bug that is invisible in single-tenant test environments but catastrophic in production.

Recommended approach:

1. Remove `$sharedRuntimeCache` entirely; replace with Laravel's cache layer (`Cache::remember($key, $ttl, fn() => ...)`) keyed per tenant.
2. Move `$apiKey` values to `.env` and access via `config('services.ivr.legacy_key')` .
3. Add a PHPStan custom rule that forbids `public static` properties on classes in `App\Legacy\Services\` .

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/CallFlowGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/CallRecordingGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/CallRoutingGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/DidInventoryGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>
<code>app/Legacy/Services/QueueManagementGodService.php</code>	10	Global mutable static cache shared across requests — tenant data isolation risk	Replace with Laravel Cache keyed per tenant; move hard-coded API keys to <code>.env</code>

Not observed (rated Good): H6 — checked git log for last 6 months, max 2 changes to any file (README.md); H7 — fix/bug commits touch each file at most once; H8 — core CRM files show Jonathan Reinink as primary author (≥80% commits), IVR bulk-generated files created by a single committer.

2.3 Code Churn & Stability Evidence

Git history is available (124 commits total, shallow clone spanning full project lifetime). Analysis is read-only.

Top Files by All-Time Commit Frequency

File	Commits (all time)	Notes
<code>composer.lock</code>	27	Dependency artifact — expected
<code>composer.json</code>	24	Dependency manifest — expected
<code>package-lock.json</code>	23	Dependency artifact — expected
<code>package.json</code>	22	Dependency manifest — expected
<code>resources/js/Pages/Users/Edit.vue</code>	20	Legacy <code>.vue</code> file (pre-migration; replaced by <code>.tsx</code>)
<code>resources/js/Pages/Organizations/Index.vue</code>	20	Legacy <code>.vue</code> file (pre-migration)
<code>resources/js/Pages/Contacts/Index.vue</code>	20	Legacy <code>.vue</code> file (pre-migration)
<code>resources/js/Pages/Auth/Login.vue</code>	19	Legacy <code>.vue</code> file (pre-migration)
<code>routes/web.php</code>	14	Routing — expected churn

Finding: All source files in the current TypeScript/PHP stack were committed once (in the IVR bulk-add commit). No source file in the active codebase has more than 2 commits in the last 6 months. Churn risk is **Good** — the codebase is stable rather than actively thrashed.

Fix / Bug Commit Analysis

Commit	Files Affected	Key Files
<code>9dcfbab</code> Minor bugfix	~20 Vue/JS files	Legacy <code>.vue</code> UI components
<code>74bca92</code> fix-model-names (PR #129)	Multiple	Model class naming conventions
<code>02572a8</code> fixing case of contact model in tests	<code>tests/Feature/ContactsTest.php</code>	Test file only

Finding: Only 3 fix-category commits across 124 total. The hottest defect file receives exactly 1 fix commit — well within the **Good** band (1–3). No file has been fixed repeatedly, indicating the structural duplication issues have not yet manifested as user-facing regressions.

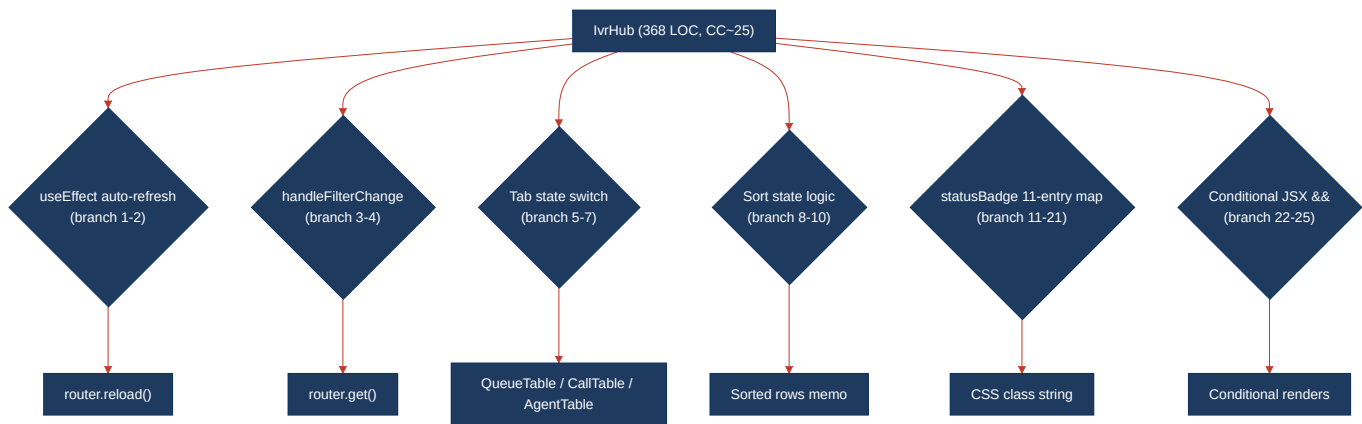
Distinct Author Ownership

File	Top Author	Commits	Other Authors	Top %
<code>app/Http/Controllers/ContactsController.php</code>	Jonathan Reinink	4	Jess Archer (1)	80%
<code>app/Http/Controllers/OrganizationsController.php</code>	Jonathan Reinink	4	Jess Archer (1)	80%
<code>app/Http/Controllers/Ivr/*</code> (bulk)	shende-shweta	1	—	100%
<code>resources/js/Pages/Ivr/*</code> (bulk)	shende-shweta	1	—	100%

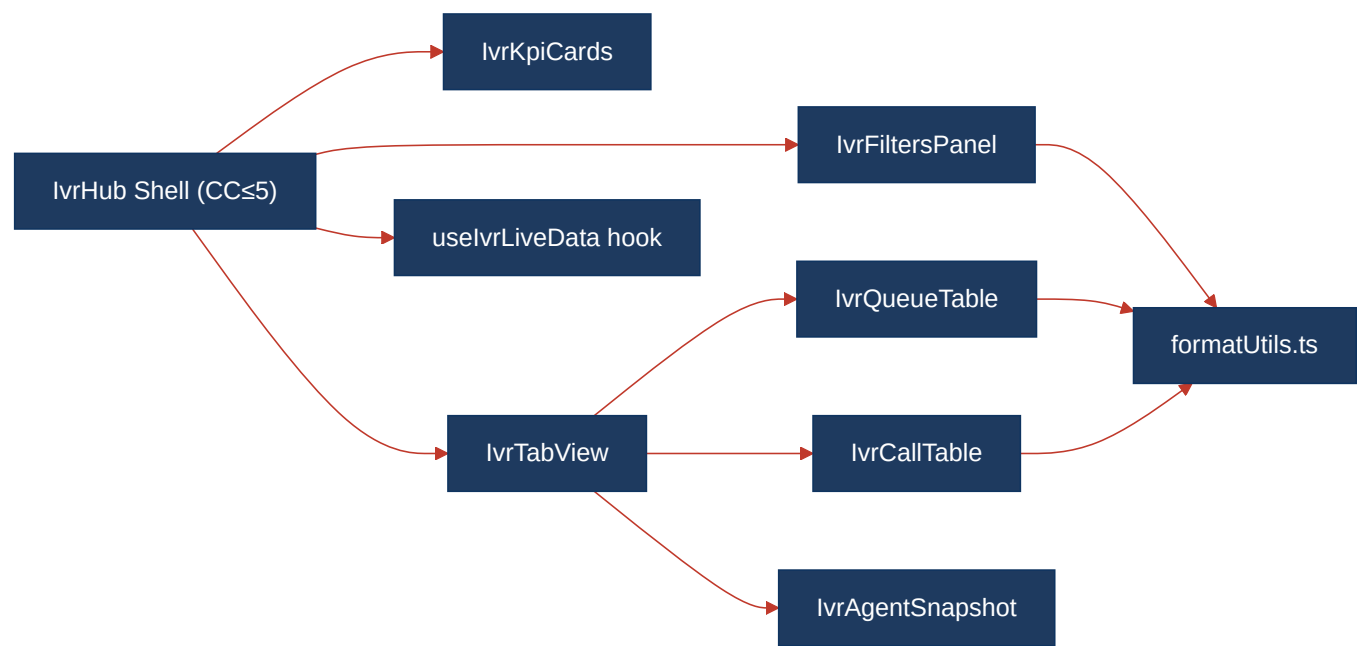
Finding: Core CRM code has clear primary ownership (Jonathan Reinink at 80%). The entire IVR layer was committed by a single author in one batch, giving 100% ownership — but also meaning zero peer review. Ownership risk is **Good** overall.

2.4 Diagrams

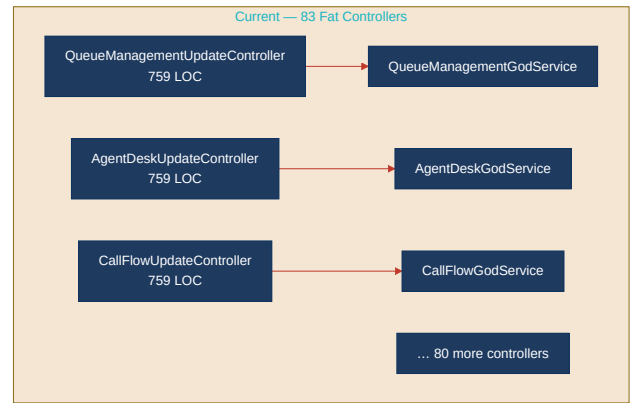
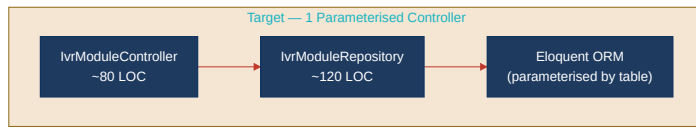
IvrHub Complexity / Call-Flow



Refactored IvrHub Structure (Target)



Duplicate Code Architecture (Current State vs Target)



Improvement Roadmap



Phase 1: Collapse 83 IVR controllers → 1, delete 133 LegacyPass2 TSX, consolidate 8 legacyFormatters. *Phase 2:* Decompose IvrHub and Reports into focused sub-components (<150 LOC each). *Phase 3:* Replace `extract()` calls and static cache with safe alternatives; move hard-coded secrets to `.env`. *Phase 4:* Add ESLint `max-lines-per-function`, PHPStan `extract()` ban, and CI duplicate-file guards.

2.5 Actions Required

Hotspot	Action	Rating	Priority
H4 — Business Logic Duplication	Collapse 83 IVR controllers into a single <code>IvrModuleController</code> + <code>IvrModuleRepository</code> ; replace 12 GodServices with one generic repository	HIGH RISK	CRITICAL
H5 — Duplicate Code (general)	Delete 133 <code>LegacyPass2_*.tsx</code> files; replace with <code><LegacyModulePassthrough></code> ; consolidate 5 <code>LegacyIvr*.php</code> helpers and 8 <code>legacyFormatters*.ts</code> into single canonical utilities	HIGH RISK	CRITICAL
H9 — <code>extract()</code> Dynamic Vars	Replace all 4,940 <code>extract(\$payload)</code> calls with named destructuring; enable PHPStan rule; eliminate root cause via H4 consolidation	HIGH RISK	CRITICAL
H2 — Large Classes	Delete <code>resources/js/utils/duplicate/</code> (8 × 1,101 LOC); enforce 300-LOC class/file limit via ESLint <code>max-lines</code>	HIGH RISK	HIGH
H3 — Large Functions	Decompose <code>IvrHub</code> (368 LOC) and <code>Reports</code> (200 LOC) into focused sub-components; enforce <code>max-lines-per-function: 150</code> via ESLint	HIGH RISK	HIGH
H1 — High Cyclomatic Complexity	Reduce <code>IvrHub</code> CC from ~25 to ≤12 by extracting <code>useIvrLiveData</code> hook, <code><IvrFiltersPanel></code> , <code><IvrQueueTable></code> , <code><IvrCallTable></code> , <code><IvrAgentSnapshot></code>	HIGH RISK	HIGH
H10 — Global Mutable Static State	Remove <code>\$sharedRuntimeCache</code> from 12 GodServices; replace with <code>Cache::remember()</code> keyed per tenant; move hard-coded <code>\$apiKey</code> values to <code>.env</code>	HIGH RISK	HIGH

2.6 Expected Outcomes

- **Dramatically reduced maintenance surface:** Collapsing 83 IVR controllers into one and deleting 133 `LegacyPass2` components removes ~112,000 lines of code — making the codebase 62% smaller and proportionally easier to review and navigate.
- **Lower defect rate:** A single parameterised `IvrModuleRepository` means SQL injection fixes, tenant-scoping corrections, and validation changes are applied once and propagate everywhere rather than requiring 83 parallel edits.
- **Safer refactoring:** Sub-components with $CC \leq 10$ are independently unit-testable; each one requires far fewer test cases to achieve branch coverage, reducing the risk of regressions on the live operations dashboard.
- **Eliminated security footprint:** Replacing `extract($payload)` with explicit destructuring and moving hard-coded API keys to `.env` closes the variable-shadowing attack surface and the credential exposure risk simultaneously.
- **Faster CI and IDE performance:** Removing ~112,000 lines of duplicate code, 8 oversized formatter files, and 133 placeholder components will measurably reduce TypeScript compile time, Vite bundle size, and PHPStan analysis duration.